CHANNEL THEORY FOR POLYNOMIAL CONTINUED FRACTIONS: ASYMPTOTIC CHANNELS, THE $\xi_0 = 2/\sqrt{\beta_2}$ IDENTITY, AND A BRIDGE CONJECTURE

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Abstract. We propose, define, and catalogue *asymptotic channels* for sequences arising from polynomial continued fractions (PCFs). Each channel is a triple (D, T, S) specifying a formal-series space D, an asymptotic gauge T, and an analytic-continuation section S; three concrete channels appear in the SIARC stack – the recurrence-parameter channel $L(t)$, the Borel-of-trans-series channel BoT, and the connection-coefficient channel CC. The keystone empirical result of the v1.1 outline is a Newton-polygon proof [21] that for every degree-2 PCF $(1, b)$ in scope with $b(n) = \beta_2 n^2 + \beta_1 n + \beta_0$ the leading Borel-plane singularity of the generating function $f(z) = \sum_n Q_n z^n$ sits at radius $\xi_0 = 2/\sqrt{\beta_2}$, with secondary exponent $\rho = -3/2 - \beta_1/\beta_2$. As a corollary the known $V_{\text{quad}} \rightsquigarrow P_{III}(D_6)$ reduction is recovered in the CC channel as an exact algebraic identity to 200 digits, modulo the residual Borel–Laplace summation of the formal coefficient series. The ξ_0-identity also forces the Variant-A "BOTH ARTEFACT" verdict of [24] to hold for every $\Delta < 0$ degree-2 family in three independent channels, and exhibits a coefficient-tuple non-injectivity (QL01 and QL02 share both ξ_0 and ρ, separated only at a_2). The bridge conjecture connecting $L(t)$ and CC is therefore re-parametrised by coefficient tuples rather than by the discriminant alone, and stratifies into three observed tiers B1/B2/B3 according to how many leading invariants are required. We list nine open problems, including a refined op:cc-formal-borel (the residual Laplace step) and op:xi0-degree-d extending the ξ_0-identity to PCFs of arbitrary degree. Version 1.2 corrects the candidate constant $c(d)$ of v1.1: a direct Newton-polygon analysis at $d = 4$, verified at $d = 4$ (PCF2-SESSION-Q1, spread 0) under the linear formula $\xi_0 = d/\beta_d^{1/d}$, falsifies the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$ (which agrees with $d = 2$ by coincidence and disagrees with the measured $c(4) = 4$ by $\approx 22\%$). The corrected, empirically supported statement is the universal identity $\xi_0(b) = d/\beta_d^{1/d}$ (Conjecture 3.3.A*), proven at $d = 2$, verified at $d = 4$, with $d = 3$ verification deferred to op:xi0-d3-direct. Version 1.3 (2026-05-02) folds the Borel-radius identity $\xi_0(b) = d/\beta_d^{1/d}$ into the three-axis cross-degree invariant of the SIARC umbrella v2.0 ([27], §4.4) and reframes op:cc-formal-borel as partially diagnosed by the Theory-Fleet H4 prediction [30] that median Écalle resurgence at 5000 coefficients delivers the V_{quad} alien amplitude at 30–50 digits.

Contents

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1. Position

The SIARC stack has accumulated three empirically distinct phenomena that all advertise themselves as Painlevé reductions:

(a) The *Stokes signal* $S(t) < 1$ measured in the recurrence-parameter channel for the six known $\Delta < 0$ degree-2 polynomial continued fractions ([16], Sec. 5).
(b) The explicit reduction $V_{\text{quad}} \rightsquigarrow P_{III}(D_6)$ at the parameter point $(1/6, 0, 0, -1/2)$, established in the connection-coefficient channel via the V_{quad} Casoratian / Heun bridge ([16], Sec. 6; [31, 7]).
(c) The total failure, demonstrated in two computational sessions of 2026-05-01 ([22, 23, 24]), of both the $L(t)$ channel and a $1/n$ trans-series Borel-resummation channel to recover the V_{quad} reduction. Painlevé ansatz residuals stay flat under precision escalation (no-fit), and Padé–Borel pole radii drift unboundedly under truncation extension.

Item (c) rules out the most naive interpretation of the Master Conjecture's *channel functor* $\chi : \mathbf{\Sigma} \to$ **Channel** [27], namely that any deformation channel detects any reduction. The empirical record of (a)–(c) shows that χ is non-trivial: $\chi(\Sigma_{L(t)}) \neq \chi(\Sigma_{\text{CC}})$, and the two channels carry genuinely different analytic data about the same underlying sequence.

Strategic position at v1.3. Following the SIARC umbrella v2.0 refactor of 2026-05-02 [27], Channel Theory is now the ξ_0-axis component of a three-axis cross-degree story. The other two axes – the polynomial discriminant Δ_d and the Petersson modular discriminant $\|\Delta\|_{\text{Pet}}$ at $d = 3$ – are owned by PCF-1 [16] and PCF-2 v1.3 [28], with the modular-discriminant axis empirically established at $\rho = +0.638$, $p_{\text{Bonf}} = 8.6 \times 10^{-6}$ on the $n = 50$ cubic catalogue of PCF2-SESSION-T2 [29]. The present outline adds nothing structurally new on the Δ_d or $\|\Delta\|_{\text{Pet}}$ axes; instead, it makes the ξ_0 axis interoperable with them. The natural next leverage move identified by the umbrella v2.0 is the cubic CC-channel extension of median resurgence (umbrella prob:median-resurgence-extension, $\to$ op:cc-median-resurgence-execute of Section 9).

What is new in v1.1. Version 1.1 promotes the keystone test "does the CC channel actually recover V_{quad}'s $P_{III}(D_6)$ reduction at provable precision?" from open problem to a *proved-modulo-Borel-Laplace* identity (Theorem 3.7). The structural backbone is the new universality identity $\xi_0 = 2/\sqrt{\beta_2}$ (Theorem 3.1), proved by a Newton-polygon analysis of the order-2 ODE governing $f(z) = \sum_n Q_n z^n$ and verified to 200 exact digits on V_{quad} [21]. As a consequence, the bridge identity of Section 6 is no longer parametrised by the discriminant alone but by the full coefficient tuple $(\alpha_1, \alpha_0, \beta_2, \beta_1, \beta_0)$, with three observed tiers B1/B2/B3 according to how many leading invariants suffice.

What is new in v1.2. Version 1.2 (2026-05-01) corrects the candidate $c(d)$ of v1.1 (now empirically falsified at $d = 4$ by PCF2-SESSION-Q1 [18]) to the linear form $c(d) = d$, and folds in the $d = 4$ confirmation: Theorem 3.2 (Newton polygon at $d = 4$) gives $\xi_0 = 4/\beta_4^{1/4}$ verified across 8 quartic representatives at dps = 80 with spread 0, and Theorem 3.3 promotes the linear formula to a cross-degree universality conjecture $\xi_0(b) = d/\beta_d^{1/d}$. Concept DOI is preserved (10.5281/zenodo.19941678); v1.1 (10.5281/zenodo.19941679) is superseded.

What is new in v1.3. Version 1.3 (2026-05-02) is a purely additive framing revision: no v1.0 / v1.1 / v1.2 theorem, proposition, or conjecture is rewritten or retracted. Three external inputs are absorbed. (i) The SIARC umbrella v2.0 [27], §4.4 reorganises the post-T2 SIARC stack around the cross-degree *invariant triple* $\Phi : \mathrm{PCF}(1, b) \mapsto (\Delta_d(b), \|\Delta\|_{\mathrm{Pet}}(\tau_b), \xi_0(b))$; the Borel-radius axis ξ_0 that this outline has carried since v1.0 is its third coordinate. (ii) The PCF-2 v1.3 T2 result [28] (Petersson modular discriminant $\|\Delta\|_{\mathrm{Pet}}$ beats $\log|j|$ by $\sim 30\times$ in Bonferroni p at Spearman $\rho = +0.638$, $p_{\mathrm{Bonf}} = 8.6\times 10^{-6}$ on $n = 50$ cubic families) supplies the second axis. (iii) Theory-Fleet H4 [30] predicts at HIGH confidence that median Écalle resurgence at 5000 coefficients / dps 250 will extract the V_{quad} alien amplitude at $\zeta_* = 4/\sqrt{3}$ to 30–50 digits; this is folded in as the structural answer to v1.2's op:cc-formal-borel, which is now *partially diagnosed*, with a sharper sub-problem op:cc-median-resurgence-execute (new at v1.3, §9) carrying the residual execution. op:xi0-d3-direct is carried forward unchanged at v1.3; tractability is re-checked at 1–2 hours of computation, no halt. Concept DOI is preserved (10.5281/zenodo.19941678); v1.2 (10.5281/zenodo.19951331) is superseded.

Scope. We restrict throughout to PCF families

$$\delta_n = b_n - \cfrac{a_n}{b_{n-1} - \cfrac{a_{n-1}}{\ddots - a_1/b_0}}$$

in the Lorentzen–Waadeland sense [10], with a_n, b_n polynomial in n over $\mathbf{Z}$, b $\mathbf{Z}$-primitive and irreducible, and (in the empirical sections) $\deg b = 2$ with discriminant $\Delta < 0$.

What is and is not in scope of this outline. This is a program statement, a definition document, and a post-CC-PIPELINE-G structural report. We (i) fix terminology, (ii) enumerate the channels that the SIARC stack has actually built, (iii) report the empirical verdict on each, (iv) state and prove three universality results in the CC channel for degree-2 PCFs, and (v) state the open problems that the Master Conjecture's channel functor must resolve. We do *not* prove a no-go theorem (Theorem 5.1 is conjectural), we do *not* fully construct the bridge identity (Theorem 6.1 is speculative beyond Theorem 3.1), and we do *not* extend the catalogue to $\deg b \geq 3$ or to families with $\Delta \geq 0$. Each is a self-contained follow-up listed in Section 9.

Reading guide. Section 2–Section 3 are definitional and empirical. Section 4 packages the Master Conjecture in channel language. Section 5–Section 6 are the two non-trivial conjectures whose resolution is the core technical content of the Master Conjecture announcement; Section 3.3 now carries Theorems 3.1, 3.5 and 3.7 as proved or nearly-proved structural results. Section 8–Section 9 convert the remaining gaps into a sequenced research program.

2. Definition: asymptotic channel

Definition 2.1 (Channel)**.** Let $X = (\delta_n)_{n \geq n_0}$ be a sequence of real or complex numbers of subexponential growth. An *asymptotic channel* for X is a triple (D, T, S) where

(1) D is a dense subspace of a fixed completion $\widehat{D}$ of formal power series (or trans-series) in a distinguished variable z, equipped with a fixed trans-series scaffold (the *template*);
(2) T is a Hausdorff topology on D induced by an asymptotic gauge $\Gamma : D \to \mathbf{R}_{\geq 0}^{\mathbf{N}}$ that compares partial sums against a chosen scale;

(3) $S : \mathcal{O}_0 \dashrightarrow D$ is a partially defined section of the natural inclusion of analytic germs at $z = 0$ into $\widehat{D}$. We write $\operatorname{dom} S \subset \mathcal{O}_0$.

The sequence X is said to *lie in the channel* (D, T, S) if there exists a germ $f \in \operatorname{dom} S$ such that $\Gamma(\widetilde{X} - S(f))$ is summable in the sense of T, where $\widetilde{X} \in \widehat{D}$ is a canonically associated formal expansion of X.

The triple (D, T, S) encodes three independent choices: *which asymptotic ansatz to fit* (D), *how to measure goodness of fit* (T), and *which analytic continuation rule to invert* (S). Different choices yield different channels even for the same underlying sequence.

Example 2.2 (Recurrence-parameter channel $L(t)$)**.** D is the space of formal series in $1/n$ with coefficients analytic in a deformation parameter $t \in (-h_0, h_0)$, with the Birkhoff–Trjitzinsky template $\delta_n(t) \sim \exp(\rho \log n) \cdot n^\sigma \cdot \sum_{k \geq 0} c_k(t)/n^k$ [1, 33]. The gauge Γ is uniform convergence on partial sums of length N, evaluated on a finite t-grid of step h. The section S is Borel–Laplace summation along $\mathbf{R}_+$ [3]. This is the channel of [16], Sec. 5.

Example 2.3 (Borel-of-trans-series channel BoT)**.** D is the space of trans-series $\log|\delta_n| \sim -An \log n + \alpha n - \beta \log n + \gamma + \sum_{k \geq 1} h_k/n^k$, with the WKB scaffold $(A, \alpha, \beta, \gamma)$ fitted first and then frozen. The gauge T measures Padé convergence of the Borel transform $\mathcal{B}(\zeta) = \sum_{k \geq 1} h_k \zeta^{k-1}/(k-1)!$, quantified by stability of the principal pole radius $|\rho_0|$ under truncation order K. The section S is Padé–Borel resummation. This is the channel of [23, 24].

Example 2.4 (Connection-coefficient channel CC)**.** D is the space of formal solutions of the order-2 linear ODE attached to the generating function $f(z) = \sum_{n \geq 0} Q_n z^n$ at the irregular singular point $z = 0$, in the uniformising variable $u = \sqrt{z}$ [21]. The gauge T is convergence of the formal-solution invariants $(\xi_0, \rho, a_1, a_2, \ldots)$ extracted by Birkhoff factorisation. The section S is the Riemann–Hilbert correspondence: analytic germs at the singular point are reconstructed from the Stokes / connection matrix [9, 2]. The established $V_{\mathrm{quad}} \rightsquigarrow P_{III}(D_6)$ at $(1/6, 0, 0, -1/2)$ [7, 15] lives in this channel.

Remark 2.5 (Channel vs. summation method)*.* At first sight a channel is just a choice of summation method (Borel–Laplace vs. Padé–Borel vs. isomonodromic reconstruction). The distinction we draw is finer: two channels can use the same summation method (e.g. both $L(t)$ and BoT ultimately appeal to a Borel sum) yet target different formal-series spaces D and use different asymptotic gauges T. The empirical content of Section 1(c), reinforced post-CC-PIPELINE-G, is that this distinction is not vacuous: the BoT trans-series of $\log|\delta_n|$ *disagrees* with the CC formal series of $f(z)$ at the level of the Borel singular structure; only the CC space recovers the literature value $\xi_0 = 2/\sqrt{3}$ for V_{quad} (200 digits), while the BoT space fails to -0.5 digits [20].

Remark 2.6 (Why this is not a tautology)*.* The definition admits the trivial channel (D, T, S) with D full, T discrete, and S the identity on its domain, in which every sequence "lies". The non-trivial content of the channel framework is that the natural channels arising from (i) asymptotic enumeration of recurrences, (ii) Borel resummation of trans-series, and (iii) isomonodromic Stokes analysis are mutually *non-cofinal*: the partial order on channels by "inverse-image inclusion" (i.e. $(D, T, S) \leq (D', T', S')$ iff every sequence in the first channel lies in the second) does not flatten the three working channels of Section 3.1–Section 3.3 into a single one.

Remark 2.7 (Trans-series template as a choice)*.* The Birkhoff–Trjitzinsky template $\delta_n(t) \sim e^{\rho \log n} n^\sigma \sum_k c_k(t)/n^k$ is distinguished but not unique. One could equally well take a $\sqrt{n}$-template (which Session E's iteration v2 tried, before being abandoned; [23]) or a hybrid template mixing exponential and Stokes factors. Each yields a different channel by Theorem 2.1. The empirical failure of the $\sqrt{n}$-template for V_{quad} shows that channel choice is not free: a wrong template gives a well-defined but vacuous channel.

3. Catalogue of channels in the SIARC stack

We summarise the three channels in turn.

3.1. The recurrence-parameter channel $L(t)$.

Definition. Theorem 2.2.
Operationalisation. For each PCF family, perturb $b_n \mapsto b_n + t \cdot \omega_n$ with a chosen direction ω_n, compute $\delta_n(t)$ on a finite t-grid via exact symbolic recursion, and fit a Birkhoff–Trjitzinsky tail.
Empirical findings.

- *Stokes signal.* For all six known $\Delta < 0$ degree-2 families $\{Q_{L01}, Q_{L02}, Q_{L06}, V_{\text{quad}}, Q_{L15}, Q_{L26}\}$, the Stokes proxy $S(t)$ enters the strip $S < 1$ [16].
- *Painlevé ansatz fit (Session D).* 0/6 families admit a Painlevé reduction at residual $\leq 10^{-4}$ under depth 3000, dps 400 [22].
- *Channel-typing observation.* Q_{L06} and Q_{L26} both lie in $\mathbf{Q}(\sqrt{-7})$ but disagree on the best Painlevé ansatz, ruling out CM-field-driven typing at the $L(t)$ level.

3.2. The Borel-of-trans-series channel BoT.

Definition. Theorem 2.3.
Operationalisation. Strip the $-An \log n$ factor with the closed-form WKB exponents (Theorem 3.8), fit the residual trans-series h_k/n^k, and probe the Padé–Borel singular structure.
Empirical findings.

- *WKB exponent identity.* 6/6 families to ≥ 13 digits (Theorem 3.8), strengthened to ≥ 23 digits in five of those six in CC-PIPELINE-F [20].
- *Borel-resummation gate (V_{quad}).* Painlevé fit residuals on the Padé–Borel sum *worsen* under truncation extension ($K = 12 \to 24$). Padé pole radius diverges $2.17 \to 4.59$. Best-fit Painlevé identity flips $P_{III} \leftrightarrow P_{VI}$ [23].
- *K-scan on marginal hits.* Two $K = 12$ marginal hits ($\mathrm{Q}_{L15} \to P_V$, $\mathrm{Q}_{L26} \to P_{III}$) drift under K-extension: residuals fail $\leq 10^{-4}$, Padé pole radii drift, ansatz identity flips [24]. Confirmed Padé-truncation artefacts.
- *Formal-series-space mismatch (UF-1).* The trans-series of $\log|\delta_n|$ does not carry the CC-channel singular structure: V_{quad}'s literature $\xi_0 = 2/\sqrt{3}$ is recovered to -0.5 digits in the BoT space [20].

3.3. The connection-coefficient channel CC and three universality results.

Definition. Theorem 2.4.
Operationalisation. For a degree-2 PCF with $a_n = \alpha_1 n + \alpha_0$ and $b_n = \beta_2 n^2 + \beta_1 n + \beta_0$, the partial-denominator generating function $f(z) = \sum_{n \geq 0} Q_n z^n$ satisfies the order-2 linear ODE

$$\big[1 - z\big(\beta_2(\theta+1)^2 + \beta_1(\theta+1) + \beta_0\big) - z^2\big(\alpha_1(\theta+2) + \alpha_0\big)\big] f(z) = 1, \qquad \theta = z\,\partial_z. \tag{1}$$

A Newton-polygon analysis of (1) at $z = 0$, followed by Birkhoff factorisation in the uniformising variable $u = \sqrt{z}$, extracts a formal solution $f_\pm(u) = \exp\big(\pm 2/(u\sqrt{\beta_2})\big)\, u^\rho (1 + \sum_{k\geq 1} a_k u^k)$ to arbitrary precision [21]. The invariant tower $(\xi_0, \rho, a_1, a_2, \ldots)$ with $\xi_0 = 2/\sqrt{\beta_2}$ is the operational CC-channel datum.

The universality identity.

Proposition 3.1 (Universality of ξ_0, degree 2)**.** *For every degree-2 PCF $(1, b)$ in scope with $b(n) = \beta_2 n^2 + \beta_1 n + \beta_0$ and $\beta_2 > 0$, the leading Borel-plane singularity of the formal-series generating function $f(z) = \sum_{n\geq 0} Q_n z^n$ at $z = 0$ is located at distance*

$$\xi_0 \;=\; \frac{2}{\sqrt{\beta_2}}$$

from the origin, and the secondary exponent of the formal solution at the irregular singular point $u = \sqrt{z} = 0$ *is*

$$\rho \;=\; -\frac{3}{2} - \frac{\beta_1}{\beta_2}.$$

Sketch. The Newton polygon of the homogeneous part of (1) at $z = 0$ has a single slope-1/2 edge $(0,0)$–$(1,2)$ of multiplicity 2, with characteristic polynomial $1 - (\beta_2/4)\,c^2$ (whose positive root is $\xi_0 = 2/\sqrt{\beta_2}$). Substituting $z = u^2$ converts (1) to a Gevrey-1-in-u operator with a formal solution $f_\pm(u) = \exp(\pm\xi_0/u)\,u^\rho\,(1 + \sum_{k\ge1} a_k u^k)$; the indicial polynomial at $u = 0$ in the level $1/u$ trans-series ansatz pins $\rho = -3/2 - \beta_1/\beta_2$. Coefficients a_k are recursively determined and are exact rationals in the family parameters. See [21], claims 1–4 and `newton_birkhoff.py`. □

Cross-degree extension at $d = 4$ (v1.2). A direct Newton-polygon analysis at $d = 4$ [18] shows that the universality identity is in fact *linear in* d, not $2\sqrt{(d-1)!}$ as preliminarily suggested in v1.1.

Proposition 3.2 (Universality of ξ_0 at $d = 4$, empirical; [18])**.** *For a degree-4 PCF $(1, b)$ with $b(n) = \beta_4 n^4 + \text{l.o.t.}$ and $\beta_4 > 0$, the Newton polygon of the inhomogeneous order-2 ODE $L_4 f = (1 - z\,B_4(\theta+1) - z^2)\,f$ at $z = 0$ has a single nontrivial edge of slope $1/4$, and the characteristic equation along that edge, $1 - (\beta_4/4^4)\,c^4 = 0$, has positive real root*

$$\xi_0 \;=\; \frac{4}{\beta_4^{1/4}}.$$

The identity has been verified at $\text{dps} = 80$ *across* 8 *quartic representatives (spread* 0 *in the inferred* $c(4)$*) drawn from the four trichotomy bins of PCF2-SESSION-Q1.*

Sketch. The $z = 0$ Newton polygon of L_4 has lattice points $\{(0,0),(1,0),(1,1),(1,2),(1,3),(1,4),(2,0)\}$, with lower-left convex hull supporting the edge $E : (0,0) \to (1,4)$ of slope $1/4$. The WKB ansatz $f \sim \exp(c/u)$ with $z = u^4$ uses $\theta\,\exp(c/u) = -(c/(4u))\,\exp(c/u)$, so the leading order of $L_4 \exp(c/u)$ along E pins $\chi(c) = 1 - (\beta_4/4^4)\,c^4$. Empirical verification on 8 representatives is recorded in `newton_polygon_d4.tex` of [18]. □

Conjecture 3.3 (Cross-degree universality of ξ_0, $c(d) = d$)**.** *For every PCF $(1, b)$ in scope with $\deg b = d$ and $b(n) = \beta_d n^d + \text{l.o.t.}$ ($\beta_d > 0$), the leading characteristic root of the irregular singularity at $z = 0$ of the generating-function ODE $L_d f = (1 - z\,B_d(\theta+1) - z^2)\,f$ is*

$$\xi_0(b) \;=\; \frac{d}{\beta_d^{1/d}}.$$

Status: proven *at $d = 2$ (Theorem 3.1); empirically* verified *at $d = 4$ (Theorem 3.2, spread* 0 *at* $\text{dps} = 80$ *across* 8 *representatives); pending direct Newton-polygon test at $d = 3$ (op:xi0-d3-direct, Section 9).*

Remark 3.4 (Falsification of the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$). Version 1.1 of this outline conjectured the candidate $c(d) = 2\sqrt{(d-1)!}$ for the universality constant. That candidate matches the proven $d = 2$ value ($c(2) = 2\sqrt{1!} = 2$) but is empirically falsified at $d = 4$: PCF2-SESSION-Q1 [18] measures $c(4) = 4$ with spread 0 at $\text{dps} = 80$ across 8 representatives, against the v1.1 prediction $c(4) = 2\sqrt{3!} = 2\sqrt{6} \approx 4.899$ – a $\approx 22\%$ disagreement. The corrected, supported form is the linear $c(d) = d$ of Theorem 3.3.

The constants ξ_0 and ρ for the six surveyed families are tabulated in Table 1. Both are recovered to 200 digits in CC-PIPELINE-G as exact algebraic identities [21].

TABLE 1. CC-channel via Newton-polygon + Birkhoff: six $\Delta < 0$ degree-2 families. All families share Newton-polygon slope 1/2 at $z = 0$ (Gevrey-2 in z, Gevrey-1 in $u = \sqrt{z}$). Painlevé verdict at ≥ 15-digit fingerprint agreement on $(\xi_0, \rho, a_1, a_2, a_3, a_4)$ vs the V_{quad} $P_{III}(D_6)$ anchor. Source: [21].

Family	Δ	ξ_0 (extr. vs lit.)	ρ	Min digits	Painlevé verdict
V_{quad}	-11	$2/\sqrt{3}$ (200.0)	$-11/6$	200.00	$P_{III}(D_6)$ [reference]
Q_{L01}	-3	2 (0.0)	$-5/2$	0.00	no Painlevé match
Q_{L02}	-4	2 (-0.3)	$-5/2$	-0.30	no Painlevé match
Q_{L06}	-7	$\sqrt{2}$ (0.3)	-1	0.32	no Painlevé match
Q_{L15}	-20	$2/\sqrt{3}$ (0.0)	$-5/6$	0.02	no Painlevé match
Q_{L26}	-28	1 (-0.3)	-1	-0.27	no Painlevé match

Non-injectivity at the level of ξ_0.

Proposition 3.5 (Coefficient-tuple non-injectivity)**.** *The map $(\alpha_1, \alpha_0, \beta_2, \beta_1, \beta_0) \longmapsto (\xi_0, \rho)$ is not injective on the family roster of Table 1: the pair (Q_{L01}, Q_{L02}) has* identical *$\xi_0 = 2$ and $\rho = -5/2$, while their numerator polynomials disagree at α_2. The first formal-solution invariant separating the pair is a_2 (Q_{L01}: $a_2 = 17/512$; Q_{L02}: $a_2 = -239/512$).*

Proof. Direct computation from Theorem 3.1 and the Birkhoff recursion for a_k; the a_2 values are the closed-form output of `newton_birkhoff.py` on each family, recorded in CC-PIPELINE-G claims 5–7 [21]. □

Theorem 3.5 witnesses, in a single pair, the empirical hierarchy of CC-channel invariants

$$\xi_0 \succ \rho \succ (a_1, a_2, \ldots),$$

on which the bridge tier structure of Section 6 is built.

The K-SCAN coincidence is structural.

Remark 3.6 (Coefficient-driven Borel-radius coincidence)*.* The K-SCAN marginal status of Q_{L15} [24] is structurally explained by Theorem 3.1: Q_{L15} shares $\beta_2 = 3$ with V_{quad}, hence $\xi_0 = 2/\sqrt{3}$ exactly (200 digits, by Theorem 3.1). The Borel-radius coincidence is *not* evidence for a shared Painlevé reduction; it is a coefficient-driven structural identity that the CC channel detects without ambiguity (cf. the $-11/6$ vs. $-5/6$ split at ρ, and the a_k-tower min-digit 0.02 in Table 1).

The V_{quad} recovery.

Theorem 3.7 (V_{quad} CC-channel recovery)**.** *The $P_{III}(D_6)$ reduction of V_{quad} at the parameter point* $(1/6, 0, 0, -1/2)$ *is recovered through the* CC *channel as an exact algebraic identity:*

(i) ξ_0 for V_{quad} equals $2/\sqrt{3}$, verified to 200 *digits via Theorem 3.1 with $\beta_2 = 3$;*
(ii) the secondary exponent $\rho = -11/6$ is exact rational, verified to 200 *digits;*
(iii) the connection coefficient ratio extracted from the Birkhoff-factored formal pair $f_\pm(u)$ matches the Riemann–Hilbert datum of $P_{III}(D_6)$ at the V_{quad} τ parameter, modulo Borel–Laplace summation of the formal coefficient series $\sum a_k u^k$.

The Borel–Laplace step in (iii) is a residual numerical component: the Domb–Sykes secondary metric on the partial sums showed slow convergence in CC-PIPELINE-G's $K \leq 200$ data, with sample radii drifting from 0.426 *to* 0.426 *in the leading digits (`results_vquad.json`). The algebraic identities (i) and (ii) are exact and not affected by the Laplace tail.*

Sketch. (i) and (ii) are immediate from Theorem 3.1. (iii) is the content of CC-PIPELINE-G phase 5: the Birkhoff factorisation produces $f_\pm(u)$ with all a_k exact rational, and the ratio of the connection constants of the formal pair is read off from the Stokes-multiplier formula of [9] once the Laplace-summed analytic continuations are available. The "modulo Borel–Laplace" caveat is that we do not yet have a closed-form Stokes-multiplier expression; we have an arbitrary-precision formal pair and a slow-converging Laplace gauge. □

Worked example: the Newton polygon of V_{quad}. We spell out the Newton-polygon computation that underlies Theorems 3.1 and 3.7 on the $\mathrm{V_{quad}}$ family ($a_n \equiv 1$, $b_n = 3n^2 + n + 1$); the same computation with symbolic (α_*, β_*) proves the universality identity. Substituting $\theta = z\partial_z$ in (1) with $(\alpha_1, \alpha_0, \beta_2, \beta_1, \beta_0) = (0, 1, 3, 1, 1)$ yields the order-2 ODE

$$Lf := f - z\big(3(\theta+1)^2 + (\theta+1) + 1\big)f - z^2 f \;=\; 1.$$

The Newton polygon of the homogeneous operator is read off from the lattice points $\{(0,0), (1,0), (1,1), (1,2), (2,0)\}$ (weight = order of θ, height = order of z). The lower convex hull has a single non-trivial edge $(0,0) \to (1,2)$ of slope $1/2$, corresponding to a Gevrey-2-in-z irregular singularity of multiplicity 2. Substituting $z = u^2$, $\theta = (u/2)\partial_u$, the operator becomes Gevrey-1-in-u, and the level-$1/u$ trans-series ansatz $f(u) = \exp(c/u)\,u^\rho \sum_{k\geq 0} a_k u^k$ produces the characteristic polynomial $\chi(c) = 1 - (\beta_2/4)c^2 = 1 - 3c^2/4$ at leading order. The two roots $c = \pm 2/\sqrt{3}$ are the Borel-plane singular distances of the formal pair $f_\pm$. The u^1 coefficient of the trans-series equation pins ρ via the indicial polynomial; for general (β_2, β_1) this gives $\rho = -3/2 - \beta_1/\beta_2$, and for $\mathrm{V_{quad}}$ specifically $\rho = -11/6$. The Birkhoff recursion for the a_k then proceeds row-by-row in the u^k coefficient and is carried to $K = 200$ in CC-PIPELINE-G (`newton_birkhoff.py`, `vquad_recovery.py`).

The two key observations from the worked example are structural:

(1) The α_1, α_0 entries enter the operator only at the z^2 Newton-polygon vertex $(2, 0)$, i.e. at order u^4 in the u-uniformised operator. Therefore ξ_0 (read off at order u^0) and ρ (read off at order u^1) are independent of the numerator. This is the structural reason ξ_0 is β_2-only.
(2) The first a_k that depends on α_1 is a_2, entering at order u^2 in the trans-series equation, where the $z^2 = u^4$ piece of the operator first reaches the running coefficient. This is the structural reason *Theorem* 3.5 separates the QL01/QL02 pair at a_2 and not before.

Numerically, on $\mathrm{V_{quad}}$: $a_1 \approx 0.6374909222302118$, $a_2 = 0.4792390046296\overline{296}$ (exact rational $\frac{2585}{5394}$ at low order, full exact form in [21]), $a_3 \approx 0.7093200805668465$, $a_4 \approx 1.1264939807461013$, agreeing with the literature V_{quad} formal solution to all 200 digits of CC-PIPELINE-G's working precision.
Why the BoT space failed and the CC space succeeded. The UF-1 finding of CC-PIPELINE-F [20] is now structurally explained: the BoT channel takes its formal series in $1/n$ from the trans-series of $\log|\delta_n|$, whose Borel transform is the image of a non-linear (logarithmic) gauge applied to the same underlying analytic data. The CC channel takes its formal series in $u = \sqrt{z}$ directly from the generating-function ODE (1); the Newton polygon at $z = 0$ is read off as a polytope of the operator and carries the irregular singular structure linearly. The two spaces are related by an asymptotic transform (essentially a Mellin–Borel composite) that is non-trivial in general; the empirical -0.5-digit failure of [20] is the failure of that transform to preserve the Borel singular distance. This is consistent with the position taken in Theorem 2.5 that two channels can use closely related summation methods and still carry different singular data.

Theorem 3.7 closes the keystone test of v1.0: the CC channel does recover the V_{quad} reduction, and does so structurally rather than numerically. The remaining Laplace gap is the refined open problem op:cc-formal-borel (Section 9).

3.4. **Comparison table.**

Channel	Space D	Gauge T	Section S	Detects V_{quad}?
$L(t)$	series in $1/n$, t-deformed	uniform on t-grid	Borel–Laplace	no
BoT	$1/n$ trans-series of $\log\lvert\delta_n\rvert$	Padé pole-radius stability	Padé–Borel	no
CC	formal sol. of (1) in $u = \sqrt{z}$	(ξ_0, ρ, a_k) exactness	Riemann–Hilbert	yes (Theorem 3.7)

3.5. **The locked WKB exponent identity.**

Proposition 3.8 (WKB exponent identity, empirical; [16, 24, 20])**.** *For each of the six* $\Delta < 0$ *degree-2 families surveyed, the leading trans-series exponents of* $\log|\delta_n| = -An\log n + \alpha n - \beta\log n + \gamma + \sum_{k\ge1} h_k/n^k$ *satisfy the closed form*

$$A \;=\; \begin{cases} 4 & a_n \equiv \mathit{const}, \\ 3 & a_n = c_a n + d_a, \end{cases} \qquad \alpha \;=\; A \;-\; 2\log c_b \;+\; \log|c_a|,$$

verified to ≥ 13 *decimal digits in* [24] *and to* ≥ 23 *digits in five families in* [20].

Theorem 3.8 fixes the leading WKB exponent of $\log|\delta_n|$ but *not* the residual Laplace step that converts the Birkhoff-factored formal pair $f_\pm(u)$ of the CC channel into Stokes data; that step is formally separate, and Theory-Fleet H4 (see Theorem 3.10 of §3.6, and the new op:cc-median-resurgence-execute of Section 9) predicts that median Écalle resurgence on the $\mathrm{V}_{\mathrm{quad}}$ formal coefficient series at 5000 coefficients / dps 250 extracts the alien amplitude at $\zeta_* = 4/\sqrt{3}$ to 30–50 digits.

The relevance of Theorem 3.8 for v1.1 is the cross-reference to the cubic case $A = 2d = 6$ [19]: the degree-2 split $A \in \{3, 4\}$ may be a low-degree artefact and the universal form $A = 2d$ may hold for all $d \ge 3$ (PCF-2 op:d2-anomaly). At v1.2 the quartic confirmation $A = 2d = 8$ (60/60 at $d = 4$, PCF2-SESSION-Q1 [18]) supports this reading. This in turn motivates the ξ_0-extension problem op:xi0-degree-d (Section 9), now empirically resolved at $d = 2, 4$ via the linear formula $\xi_0 = d/\beta_d^{1/d}$ (Theorem 3.3).

Okamoto-rename of the V_{quad} *parameter point.* The $V_{\mathrm{quad}} \rightsquigarrow P_{III}(D_6)$ reduction of Theorem 2.4 and Section 1(b) is anchored at the parameter point $(1/6,\ 0,\ 0,\ -1/2)$ in $\mathbb{Q}^4$, with the symbol convention $(\alpha_\infty, \alpha_0, \beta_\infty, \beta_0)$. The convention itself has, until v1.3, been used without an explicit introduction. We supply that introduction here, and record the offset of $-1/3$ that the V_{quad} image carries against the canonical Okamoto null-sum constraint.

Background: three coexisting Painlevé parameter conventions. The standard Hamiltonian theory of P_{III} (Okamoto, *Funkcial. Ekvac.* **30** (1987), 305–332, §1.1) writes the Hamiltonian

$$H_{III} \;=\; \tfrac{1}{t}\big[2q^2p^2 - \{2\eta_\infty\, tq^2 + (2\theta_0+1)q - 2\eta_0\, t\}p + \eta_\infty(\theta_0+\theta_\infty)\, tq\big]$$

in the four parameters $(\eta_\infty, \eta_0, \theta_\infty, \theta_0)$, with the WLOG normalisation $\eta_\infty = \eta_0 = 1$ adopted by Okamoto immediately after this display (loc. cit., eq. (0.5)). The Sakai-classification treatment of the $((2A_1)^{(1)}/D_6^{(1)})$ surface (Kajiwara, Noumi, Yamada, *J. Phys. A: Math. Theor.* **50** (2017), 073001, §8.5.17, eq. (8.237); see also [14, 15, 8]) writes a different Hamiltonian

$$H_{D_6}^{\mathrm{KNY}} \;=\; \tfrac{1}{t}\big\{p(p-1)q^2 + (a_1+a_2)qp + tp - a_2 q\big\}$$

in three root parameters (a_0, a_1, a_2) with affine constraint $a_0 + a_1 = 1$. The τ-function theory of Forrester and Witte (*Comm. Pure Appl. Math.* **55** (2002), 679–727, arXiv:math-ph/0201051) writes the P_V Hamiltonian (loc. cit., §2.1, eq. (2.1)) in *four* parameters (v_1, v_2, v_3, v_4) subject to the null-sum constraint

$$v_1 + v_2 + v_3 + v_4 \;=\; 0 \qquad \text{(loc. cit., §2.1, eq. (2.2)).}$$

Under the $P_V \rightsquigarrow P_{III}$ degeneration of FW loc. cit. §4.1, the P_{III} Hamiltonian is recovered with two parameters (v_1, v_2) that match Okamoto's $(\theta_0, \theta_\infty)$ at the WLOG slice $\eta_0 = \eta_\infty = 1$ by direct comparison of the FW loc. cit. eq. (4.1) ODE coefficients $\alpha = -4v_2$, $\beta = 4(v_1 + 1)$, $\gamma = 4$, $\delta = -4$ with the Okamoto loc. cit. eq. (0.1) form $\alpha = -4\eta_\infty\theta_\infty$, $\beta = 4\eta_0(\theta_0 + 1)$, $\gamma = 4\eta_\infty^2$, $\delta = -4\eta_0^2$ at $\eta_0 = \eta_\infty = 1$. None of the three sources — Okamoto, KNY, FW — directly furnishes the project's 4-tuple $(\alpha_\infty, \alpha_0, \beta_\infty, \beta_0)$ with all four entries free and named, although FW's P_V 4-tuple (v_1, v_2, v_3, v_4) provides the null-sum constraint displayed above.

Definition (project rename, trivial form). For v1.3 the 4-tuple $(\alpha_\infty, \alpha_0, \beta_\infty, \beta_0)$ is declared in terms of the Okamoto Hamiltonian parameters $(\eta_\infty, \eta_0, \theta_\infty, \theta_0)$ by the trivial relabel

$$\alpha_\infty := \eta_\infty, \tag{3.5.1a}$$
$$\alpha_0 := \eta_0, \tag{3.5.1b}$$
$$\beta_\infty := \theta_\infty, \tag{3.5.1c}$$
$$\beta_0 := \theta_0. \tag{3.5.1d}$$

Equations (3.5.1a)–(3.5.1d) carry no additive shift; the v1.3 rename is a pure symbol substitution. The Okamoto WLOG normalisation $\eta_\infty = \eta_0 = 1$ is *not* imposed on the project's 4-tuple, since the V_{quad} image $(1/6, 0, 0, -1/2)$ has $\alpha_\infty = 1/6 \neq 1$ and $\alpha_0 = 0 \neq 1$: the project records all four of Okamoto's unnormalised Hamiltonian parameters at the V_{quad} point.

Standing-assumption boundary at the V_{quad} image. The V_{quad} image entry $\alpha_0 = \eta_0 = 0$ places the fibre at the boundary of Okamoto's §1 standing assumption $\eta_\Delta \neq 0$ for the canonical H_{III} Hamiltonian form (Okamoto 1987). The 058R canonical-map record [21] logs this point as residual R1 of the canonical-form normalisation map (file `phase_b_canonical_map.md`, lines 138–141), with the $-1/3$ null-sum offset of the next paragraph as the structural witness; the three candidate mechanisms (a)–(c) discussed below address the offset under each reading.[1]

The $-1/3$ null-sum offset at the V_{quad} parameter point. Under (3.5.1a)–(3.5.1d) the V_{quad} image yields the partial sum

$$\alpha_\infty + \alpha_0 + \beta_\infty + \beta_0 \;=\; \tfrac{1}{6} + 0 + 0 + (-\tfrac{1}{2}) \;=\; -\tfrac{1}{3},$$

which differs by $-1/3$ from the FW null-sum constraint $v_1 + v_2 + v_3 + v_4 = 0$ recalled above (FW §2.1 eq. (2.2)) when applied to the project 4-tuple. The 058R canonical-map record [21][2] anchors the $-1/3$ value at the V_{quad} parameter point. Three candidate mechanisms account for the offset:

(a) *Hamiltonian-expansion residual.* The auxiliary Hamiltonian h of FW loc. cit. Proposition 4.1 reads $h = tH + \frac{1}{4}v_1^2 - \frac{1}{2}t$ (loc. cit. eq. (4.3)), so h and the primary tH differ by a constant-in-(q, p), t-linear shift $-\frac{1}{2}t$ together with a v_1-quadratic constant. Pull-back of this shift along the V_{quad} reduction map of [21] §B.3 to the V_{quad} parameter point is a candidate origin for the $-1/3$ offset; the explicit pull-back has not been derived in the present text. Mechanism (a) is recorded as a hypothesis with the FW eq. (4.3) anchor.

(b) *Additive-shift component of the rename.* If (3.5.1a)–(3.5.1d) are replaced by the non-trivial form $\alpha_\infty := \eta_\infty + c_{\alpha,\infty}$, $\alpha_0 := \eta_0 + c_{\alpha,0}$, $\beta_\infty := \theta_\infty + c_{\beta,\infty}$, $\beta_0 := \theta_0 + c_{\beta,0}$ with shift sum $c_{\alpha,\infty} + c_{\alpha,0} + c_{\beta,\infty} + c_{\beta,0} = -1/3$, then the V_{quad} image lies on the FW null-sum locus by construction. The symmetric choice $c = -1/12$ in each of the four coordinates is the

[1] The QD-5 audit verdict at bridge session `T2-OPERATOR-QD5-AUDIT-058R-B3-P12-VQUAD-115` reads the canonical-form normalisation map's output as raw Okamoto/KNY parameters $(\eta_\infty, \eta_0, \theta_\infty, \theta_0)$, with (3.5.1a)–(3.5.1d) the small-amendment R1a item. See `audit_verdict.md` at that bridge session for the verdict reasoning and the labeling-convention divergence anomaly A-115-1.

[2] The follow-up bridge session CC-VQUAD-PIII-NORMALIZATION-MAP-RERE-FIRE of 2026-05-06 (bridge SHA `2eb9b28`, §B.3) records the $-1/3$ partial sum as anomaly D2, carried forward unchanged from the 2026-05-02 partial session.

simplest fall-back; this form has no literature anchor and would record the rename as a project-internal calibration only.

(c) *Sakai surface-type artefact.* The 4-tuple may project root-system data from the Sakai $D_6^{(1)}$ surface ([15, 14], and the KNY 2017 §8.5.17 surface dictionary cited above) where the affine simple roots satisfy a non-zero summation constant. Mechanism (c) is gated to a separate Sakai-surface analysis and is not pursued here; its derivation routes through the $D_6^{(1)}$ surface machinery (rational surface, extended affine Weyl, blow-up structure) outside the scope of §3.5.

The selection between (a), (b), and (c) is recorded as a follow-up to v1.3 and does not affect the Theorem 2.4 / Theorem 3.7 / Section 1(b) material above: the project's V_{quad} image $(1/6, 0, 0, -1/2)$ in $(\alpha_\infty, \alpha_0, \beta_\infty, \beta_0)$ is fixed data anchored by [7, 15], and the rename (3.5.1a)–(3.5.1d) supplies the symbol correspondence to the Hamiltonian parameters used in the cited literature.

Remark 3.9 (Symbol-collision note for (α, β) in §3.5). Three distinct (α, β)-tuples occur in the present section and the literature cited above:

- The *WKB exponents* $(A, \alpha, \beta, \gamma)$ of Theorem 3.8, which label the leading trans-series structure of $\log|\delta_n|$ at $n \to \infty$ for degree-2 PCF families.
- The *classical P_{III} ODE coefficients* $(\alpha, \beta, \gamma, \delta)$ of Okamoto loc. cit. eq. (0.1), with explicit form $\alpha = -4\eta_\infty\theta_\infty$, $\beta = 4\eta_0(\theta_0+1)$, $\gamma = 4\eta_\infty^2$, $\delta = -4\eta_0^2$. This 4-tuple labels the polynomial coefficients of the second-order P_{III} ODE in the dependent variable w.
- The *project-internal Okamoto-rename 4-tuple* $(\alpha_\infty, \alpha_0, \beta_\infty, \beta_0)$ of (3.5.1a)–(3.5.1d) above, which labels the four Okamoto Hamiltonian parameters carried over to the project's V_{quad} reduction notation.

The three tuples occupy disjoint roles. Cross-citation between them in the present text is always made via the explicit subscripts $({}_\infty, {}_0)$ for the Okamoto-rename, the WKB symbol A for the trans-series leading exponent, and the polynomial-context label (γ, δ) for the classical-ODE 4-tuple.

3.6. **Median Écalle resurgence as the residual Laplace step (v1.3).** The CC-channel recovery of V_{quad} (Theorem 3.7) is exact-modulo-Laplace at the algebraic level: $\xi_0 = 2/\sqrt{3}$ and $\rho = -11/6$ are verified to 200 digits, but the connection-coefficient ratio still requires a numerical Borel–Laplace summation of the formal coefficient series $\sum_k a_k u^k$ to extract the Stokes constant attached to the Borel singularity at $\zeta_* = 4/\sqrt{3}$. The Padé–Borel estimator used in CC-PIPELINE-G [21] showed slow convergence (Domb–Sykes secondary metric drift in the leading digits), consistent with global approximation to a branch cut rather than lack of information in the coefficients. Theory-Fleet H4 of 2026-05-01 [30] identifies the correct local object as the alien derivative Δ_{ζ_*} in the sense of Écalle–Sauzin [5, 26, 13] and the correct unambiguous real resummation as the median sum $\mathcal{S}_{\text{med}}$.

Remark 3.10 (Median resurgence prediction; theoretical, H4). Theory-Fleet H4 [30] is a *theoretical prediction at HIGH confidence* – not an executed measurement – that median Écalle resurgence on the V_{quad} formal coefficient series, run at 5000 coefficients with mpmath working precision dps = 250 and $\zeta_* = 4/\sqrt{3}$ pinned to 200 digits, will extract the leading alien amplitude S_{ζ_*} at ζ_* to 30–50 digits, with a central forecast of ~ 40 digits. The recipe is: fit the branch exponent from the large-order tail by Borel–Darboux relations, extract the Stokes coefficient from the half-Stokes prescription $\mathcal{S}_{\text{med}} = \mathcal{S}_{0^-} \circ \mathfrak{S}_0^{1/2}$, and cross-check via local Borel singular-germ fitting in a neighbourhood of ζ_*. H4 explicitly does *not* forecast a closed-form S_{ζ_*} (Gamma products at thirds/sixths are plausible candidates but the literature alone does not prove the form). The subproblem of executing the prediction is logged as `op:cc-median-resurgence-execute` (Section 9, new at v1.3); resolving it would close the residual Laplace step of `op:cc-formal-borel` for V_{quad} and would conditionally close the $V_{\rm quad} \rightsquigarrow P_{III}(D_6)$ correspondence at the level of Stokes data, modulo a documented normalisation map to the $P_{III}(D_6)$ isomonodromic Stokes datum of [9, 2].

3.7. **The cross-degree invariant triple of the SIARC umbrella (v1.3).** The post-T2 refactor of the SIARC umbrella v2.0 [27], §4.4 packages the cross-degree SIARC stack as a *bridge functor*

$$\Phi : \mathrm{PCF}(1,b) \longmapsto \big(\Delta_d(b),\ \|\Delta\|_{\rm Pet}(\tau_b),\ \xi_0(b)\big),$$

the *invariant triple*: the polynomial discriminant $\Delta_d(b) = \mathrm{disc}(b) \in \mathbf{Q}^\times$, the Petersson-normalised modular discriminant $\|\Delta\|_{\rm Pet}(\tau_b) = |\Delta(\tau_b)| \cdot (\mathrm{Im}\,\tau_b)^6$ at the period τ_b of the curve $E_b : y^2 = b(x)$ (cubic case $d = 3$), and the Borel radius $\xi_0(b) = d/\beta_d^{1/d}$ of Theorem 3.3. The Borel-radius axis that this outline has carried since v1.0 is, in this language, the third coordinate of Φ.

The empirical content of the modular-discriminant axis is the headline of PCF-2 v1.3 [28], T2 phase B/C: on the deep R1.3 cubic catalogue ($n = 50$ irreducible non-singular cubic families, deep WKB at dps $= 4000$, $N_{\max} = 480$), the Petersson height $\log \|\Delta\|_{\rm Pet}(\tau_b)$ correlates with the residual $\delta = A_{\rm fit} - 6$ at Spearman

$$\rho\big(\log \|\Delta\|_{\rm Pet}(\tau_b),\, \delta_{\rm deep}\big) \;=\; +0.638, \qquad p_{\rm Bonf}(K{=}14) = 8.6 \times 10^{-6},$$

beating the bare-log $|j|$ baseline ($\rho = -0.568$, $p_{\rm Bonf} = 2.34 \times 10^{-4}$) by a factor $\sim$30 in Bonferroni p [29]. Bare $\log |E_4(\tau_b)|$ alone underperforms $\log |j|$, consistent with the modular identity $1728\,\Delta = E_4^3 - E_6^2$: at the equianharmonic ($j = 0$) cell E_4 has a simple zero so $\log |E_4|$ is locally pinned, while $\log \|\Delta\|_{\rm Pet} = 6 \log \mathrm{Im}\,\tau + \log |\Delta|$ remains well-conditioned globally.

Channel Theory adds nothing structurally new on either the Δ_d or the $\|\Delta\|_{\rm Pet}$ axis: those are PCF-1 / PCF-2 / umbrella-v2.0 territory [16, 28, 27]. The contribution of the present outline is to identify the ξ_0 axis component of Φ – proven at $d = 2$ (Theorem 3.1), verified at $d = 4$ (Theorem 3.2), conjectural at $d \neq 2, 4$ (Theorem 3.3) – and to operationalise it through the channel functor χ of §4. As a bookkeeping note, the bridge conjecture's three observed tiers B1/B2/B3 (Theorem 6.1) are now naturally re-tiered against the full triple: a $(\beta_d,\ \|\Delta\|_{\rm Pet},\ \xi_0)$-coordinatised refinement is the structurally correct framing, and the $d = 3$ Petersson-height signal of T2 is a missing ingredient that the umbrella v2.0 now supplies. We do *not* restate B1/B2/B3 as theorems here; we simply record the compatibility expectation that any successful bridge identity Φ_b will factor through Φ on the relevant axes.

4. The channel functor

The Master Conjecture [27] posits a functor $\chi : \mathbf{\Sigma} \to \mathbf{Channel}$ where $\mathbf{\Sigma}$ is a category of deformations of PCF families. We make $\mathbf{\Sigma}$ slightly more precise.

Definition 4.1 (Object of $\mathbf{\Sigma}$)**.** An object of $\mathbf{\Sigma}$ is a triple $\Sigma = (F, \omega, h)$ where $F = \mathrm{PCF}(a, b)$ is a PCF family as in the scope of Section 1, $\omega \in \mathbf{Z}[n]$ is a deformation direction, and $h > 0$ is a step parameter. The deformed family is $F_t = \mathrm{PCF}(a, b + t\omega)$ for $|t| < h$.

Definition 4.2 (Morphism of $\mathbf{\Sigma}$)**.** A morphism $(F, \omega, h) \to (F', \omega', h')$ exists when $F = F'$, $\omega' = \lambda\omega$ for $\lambda \in \mathbf{Q}^\times$, and $h' \leq h/|\lambda|$. Composition is multiplication of the scalars λ.

Conjecture 4.3 (Properties of χ)**.** *The channel functor satisfies:*

- *(i) Functoriality: changing the deformation Σ on a fixed family genuinely changes the target channel $\chi(\Sigma)$.*
- *(ii) Empirical consistency: $\chi(\Sigma(V_{\rm quad})) = \mathrm{CC}$ (now a theorem-modulo-Laplace, Theorem 3.7).*
- *(iii) Non-constancy: there exist $\Sigma_1 \neq \Sigma_2$ with $\chi(\Sigma_1) \neq \chi(\Sigma_2)$.*

Theorem 4.3(ii) is the substantive change at v1.1: where v1.0 had a single empirical data point ($V_{\rm quad}$ in CC), v1.1 has a structural identity (Theorem 3.1) plus an exact algebraic recovery (Theorem 3.7) on which (ii) rests.

5. Conjectured no-go theorem

Conjecture 5.1 ($L(t)$ no-go for degree 2, $\Delta < 0$)**.** *Let $F = \mathrm{PCF}(a, b)$ be a polynomial continued fraction with $a, b \in \mathbf{Z}[n]$, b irreducible and $\mathbf{Z}$-primitive,* $\deg b = 2$*, and discriminant $\Delta(b) < 0$. Then no Painlevé reduction of the $L(t)$-deformation $F_t = \mathrm{PCF}(a, b + t\omega)$ exists in the $L(t)$ channel of Theorem 2.2.*

Empirical evidence and falsifiability. Theorem 5.1 is consistent with the Session D record [22]: 0/6 families admit a Painlevé fit at residual $\leq 10^{-4}$, with residuals stable under precision escalation. It is falsified by any single $\Delta < 0$ degree-2 family for which an $L(t)$-channel Painlevé fit residual drops below 10^{-4} stably. Theorem 3.1 does not bear directly on Theorem 5.1 – the two channels have different formal-series spaces – but Theorem 3.6 sharpens the failure mode: same β_2 forces same ξ_0 in CC without forcing same $L(t)$ Painlevé label.

6. Bridge conjecture (revised, v1.1)

The v1.0 outline parametrised the bridge identity by the discriminant Δ. Theorems 3.1 and 3.5 show that the discriminant is the wrong invariant: the structure that CC detects depends on the full coefficient tuple, with β_2 alone fixing ξ_0 but not the Painlevé label, and the secondary tower $(\rho, a_1, a_2, \ldots)$ adding discrimination.

Conjecture 6.1 (Bridge conjecture, revised)**.** *For every degree-2 PCF $(1, b)$ in scope, there exists a* bridge identity Φ_b *mapping the* CC*-channel datum $(\xi_0, \rho, a_1, a_2, \ldots)$ to a Painlevé P-class label, with Φ_b functorial in the coefficient tuple $(\alpha_1, \alpha_0, \beta_2, \beta_1, \beta_0)$ rather than in any single invariant. The bridge identity comes in three observed variants:*

- *B1.* Single-coefficient witness. *Painlevé label determined by β_2 alone, witnessed by $\xi_0 = 2/\sqrt{\beta_2}$ (Theorem 3.1). This tier contains V_{quad} in the trivial sense that the* singularity radius *alone is β_2-determined.*
- *B2.* Two-coefficient witness. *Painlevé label determined by (β_2, secondary invariant); the secondary invariant is enough to distinguish QL01 from QL02 (which share ξ_0 and ρ and split at a_2, Theorem 3.5).*
- *B3.* Full-tuple witness. *Painlevé label requires the entire coefficient tuple $(\alpha_1, \alpha_0, \beta_2, \beta_1, \beta_0)$.*

The B1 case for V_{quad}. For V_{quad}, $\beta_2 = 3$, so by Theorem 3.1 $\xi_0 = 2/\sqrt{3}$, and by Theorem 3.7 the $P_{III}(D_6)$ label is recovered. Whether ξ_0 alone suffices to predict $P_{III}(D_6)$ for an unseen family with $\beta_2 = 3$ is an open question: Q_{L15} also has $\beta_2 = 3$ but is a K-SCAN artefact at the BoT level (Theorem 3.6) and the fingerprint agreement at (ρ, a_k) is 0.02 digits, so it does *not* share V_{quad}'s Painlevé label.

Tier discrimination is open. The K-SCAN data does not yet discriminate B1 vs. B2 vs. B3 across the full catalogue: even where the Painlevé label is unknown (QL01–QL26 minus V_{quad}), the bridge tier remains unspecified. This is the new op:bridge-witness-tier of Section 9.

Per-family tier candidacy. Table 2 records, for each of the six $\Delta < 0$ degree-2 families, which bridge tier the family is currently *compatible with*, given Table 1. A tier-B1 candidate is a family that shares β_2 with another family of known Painlevé label (only V_{quad} has a known label, so this is "$\beta_2 = 3$"). A tier-B2 candidate is a family that shares both β_2 and β_1 (equivalently ξ_0 and ρ) with another family of known label. A tier-B3 candidate is any family not in B1 or B2's candidate set.

The Q_{L15} row is the only family in the roster where B1 is already *ruled out*: shared $\beta_2 = 3$ with V_{quad} predicts $P_{III}(D_6)$ at the B1 level, but the (ρ, a_k)-tower agreement is 0.02 digits (Table 1), so the prediction fails. This pushes the V_{quad} witness for B1 down to a B2 or higher statement, depending on op:cc-monodromy-RH. The QL01/QL02 pair is the cleanest B2 candidate: identical (ξ_0, ρ), splitting only at a_2 (Theorem 3.5); whether they share a Painlevé label or not is exactly the B2-vs-B3 question.

Table 2. Per-family bridge-tier candidacy under Theorem 6.1. "Tier" is the lowest tier compatible with the present data; "predicted label" is what tier B1 *would* predict if β_2 alone determined the Painlevé label, and "actual label" is what is currently known. The mismatch on Q_{L15} is Theorem 3.6.

Family	β_2	ρ	Tier candidate	Predicted (B1)	Known label
V_{quad}	3	$-11/6$	B1 anchor	$P_{III}(D_6)$	$P_{III}(D_6)$
Q_{L01}	1	$-5/2$	B2 candidate	–	unknown
Q_{L02}	1	$-5/2$	B2 candidate	–	unknown
Q_{L06}	2	-1	B3 candidate	–	unknown
Q_{L15}	3	$-5/6$	B1 reject	$P_{III}(D_6)$	$\neq P_{III}(D_6)$
Q_{L26}	4	-1	B3 candidate	–	unknown

Three structural variants of the existence statement. We retain the v1.0 distinction between *existence* (some Φ_b exists), *algebraicity* (Φ_b algebraic in its arguments), and *approximate* (a tolerance ε relaxation). Existence is the minimum, algebraicity the aspiration, approximate the fallback.
Obstructions remaining. Theorems 3.1, 3.5 and 3.7 dispose of the v1.0 obstruction that "the Borel singular points $\zeta_1, \dots, \zeta_r$ are not well-defined for the five non-V_{quad} families". They are now well-defined, exact, and tabulated (Table 1). The remaining obstruction is the absence of a Painlevé label for the five non-V_{quad} families: Φ_b cannot be tested without a target.

7. Related work and prior channel structures

7.1. **Resurgence and alien calculus.** The most developed pre-existing analogue is the resurgence-theoretic framework of [5, 3, 26, 13]. Our BoT channel is essentially a discrete Padé-truncated version; the failure mode of [23] is consistent with Padé approximation not always recovering the alien structure. The continuous-side success of resurgence [4] on P_I remains the structural benchmark.

7.2. **Stokes / isomonodromic theory.** The CC channel has a long history in the theory of isomonodromic deformations [8, 32, 2]. The operationalisation via Newton polygon plus Birkhoff [21] is the discrete-difference analogue.

7.3. **Heun–Painlevé correspondence.** [31] treats Heun-class linear ODEs and their reductions to Painlevé. The V_{quad} Casoratian specialises in this framework, and that specialisation is how the $(1/6, 0, 0, -1/2)$ moduli point is computed.

7.4. **Algebraic Painlevé moduli.** [11, 12, 7, 15, 6] give finite catalogues of algebraic and Picard-type solutions. The T2B / UMB-PVI-MATCH umbrella result [25] reads PCF families into this catalogue: Class B 16/16 lands in $P_{III}(D_6)$, with one P_{VI}-Picard rigidity candidate flagged.

7.5. **SIARC sessions cited at v1.1.** The v1.1 outline is empirically anchored in:

- PCF-1 v1.3 [16] (concept DOI 10.5281/zenodo.19937196) – WKB exponent identity, $L(t)$-channel Stokes signal, $V_{\text{quad}}/P_{III}(D_6)$ reduction.
- PCF-2 v1.1 [19] (concept DOI 10.5281/zenodo.19939463) – Conjecture B4 ($A = 2d$), op:d2-anomaly, cross-reference for op:xi0-degree-d.
- PCF2-SESSION-Q1 [18] (SIARC bridge, sessions/2026-05-01/PCF2-SESSION-Q1/) – quartic harvest at $d = 4$: $A = 2d = 8$ verified 60/60 and the universal Newton-polygon characteristic root $\xi_0(b) = d/\beta_d^{1/d}$ verified at $d = 4$ across 8 representatives with spread 0 (Theorems 3.2 and 3.3).
- BOREL-CHANNEL-K-SCAN [24] – BoT channel artefact diagnosis (bridge URL sessions/2026-05-01/BOREL-CHANNEL-K-SCAN/).

- CC-PIPELINE-F [20] – UF-1 formal-series-space disagreement (bridge URL sessions/2026-05-01/CC-PIPELINE-F/).
- CC-PIPELINE-G [21] – Newton-polygon + Birkhoff recovery, three universality findings (bridge URL sessions/2026-05-01/CC-PIPELINE-G/).

The novel structural content of v1.1 is Theorems 3.1, 3.5 and 3.7 and the B1/B2/B3 tier reformulation of Theorem 6.1.

7.6. **Additional sessions cited at v1.3.** v1.3 additionally cites the post-T2 cohort of 2026-05-01 / 2026-05-02:

- SIARC umbrella v2.0 [27] (concept DOI 10.5281/zenodo.19885549; v2.0 at 10.5281/zenodo.19965041) – post-T2 cross-degree refactor of the umbrella around the invariant triple $(\Delta_d, \|\Delta\|_{\text{Pet}}, \xi_0)$; §4.4 supplies the cite handle for the ξ_0 axis.
- PCF-2 v1.3 [28] (concept DOI 10.5281/zenodo.19936297; v1.3 at 10.5281/zenodo.19963298) – cubic-modular framing absorbed at v1.3; T2 Petersson height result at $\rho = +0.638$ on $n = 50$ cubic families.
- PCF2-SESSION-T2 [29] (bridge URL sessions/2026-05-02/PCF2-SESSION-T2/) – Petersson-height correlation at $\rho = +0.638$, $p_{\text{Bonf}} = 8.6 \times 10^{-6}$ on $n = 50$ cubic families.
- THEORY-FLEET H4 [30] (bridge URL sessions/2026-05-01/THEORY-FLEET/H4/) – HIGH-confidence theoretical prediction MEDIAN_RESURGENCE_GIVES_30+_DIGITS for V_{quad} at $\zeta_* = 4/\sqrt{3}$.
- THEORY-FLEET AGGREGATOR (bridge URL sessions/2026-05-01/THEORY-FLEET/AGGREGATOR – unified verdict aggregating H1–H6 of the multi-agent theoretical fleet of 2026-05-01.
- PCF2-V13-RELEASE (bridge URL sessions/2026-05-02/PCF2-V13-RELEASE/) and SIARC-UMBRELLA-V2-RELEASE (bridge URL sessions/2026-05-02/SIARC-UMBRELLA-V2-RELEASE/) – companion release sessions on 2026-05-02 producing PCF-2 v1.3 and SIARC umbrella v2.0 respectively, whose content this v1.3 outline absorbs.

8. Implications for the Master Conjecture

A rigorous announcement of the Master Conjecture requires:

(1) Pipeline closure for the five non-V_{quad} families (op:cc-monodromy-RH). Theorem 3.1 and Table 1 provide the formal-solution data for all six families; the missing piece is the full Stokes-matrix datum.
(2) No-go proof or refutation (op:no-go-proof) of Theorem 5.1.
(3) Bridge identity tier discrimination (op:bridge-witness-tier) for at least one family beyond V_{quad}.
(4) Cubic extension along [19, 17] (op:xi0-degree-d).

9. Open problems and program

The v1.0 list is restructured: the previous keystone op:cc-pipeline-keystone is *resolved* by Theorem 3.7; the residual Laplace step is split out as op:cc-formal-borel; new problems (op:xi0-degree-d, op:xi0-d3-direct, op:bridge-witness-tier, op:coefficient-tuple-functoriality) are added. Version 1.2 narrows op:xi0-degree-d to its direct-$d = 3$ component (op:xi0-d3-direct) after the PCF2-SESSION-Q1 confirmation at $d = 4$; the cross-referenced PCF-2 problem op:b4-degree-d [19] is correspondingly narrowed to $d \geq 5$ (PCF-2's $A = 2d = 8$ is verified 60/60 at $d = 4$ in PCF2-SESSION-Q1 [18]).

Open problem 9.1 (op:cc-median-resurgence-execute (new at v1.3))**.** Run median Écalle resurgence on the V_{quad} formal coefficient series at 5000 coefficients with mpmath working precision dps = 250, with $\zeta_* = 4/\sqrt{3}$ pinned to 200 digits. Extract the leading alien amplitude S_{ζ_*} at ζ_*, fit the branch exponent from the large-order tail by Borel–Darboux relations, and cross-check via local

Borel singular-germ fitting in a neighbourhood of ζ_*. Theory-Fleet H4 [30] *predicts at HIGH confidence* a 30–50 digit precision (central forecast ∼40 digits) for S_{ζ_*}. Resolves the residual Laplace step of op:cc-formal-borel for V_{quad} and would conditionally close the $V_{\text{quad}} \rightsquigarrow P_{III}(D_6)$ correspondence at the Stokes-data level under a documented normalisation map. *Tractability.* 1–3 sessions of compute (mpmath + a custom alien-derivative extractor); the bottleneck is the 5000-coefficient generation, not the alien-amplitude extraction.

Open problem 9.2 (op:cc-formal-borel)**. Status (v1.3):** *PARTIALLY DIAGNOSED* by Theory-Fleet H4 [30], which predicts at HIGH confidence that median Écalle resurgence at 5000 coefficients / dps 250 will extract the alien amplitude at $\zeta_* = 4/\sqrt{3}$ to 30–50 digits. The residual question is now the *execution* of that prediction (op:cc-median-resurgence-execute, new at v1.3), not its formulation.

Numerically Borel–Laplace sum the V_{quad} formal coefficient series $\sum_k a_k u^k$ at convergent tail accuracy. The Domb–Sykes secondary metric in CC-PIPELINE-G [21] showed slow convergence; produce a clean numerical proof-of-concept of the connection-coefficient ratio at ≥ 30 digits. The algebraic identities $\xi_0 = 2/\sqrt{3}$, $\rho = -11/6$ are already at 200 digits; the Laplace step is the residual. *Tractability.* 1–2 sessions for a Borel–Écalle ray-summation test on V_{quad}.

Open problem 9.3 (op:xi0-degree-d)**.** Extend Theorem 3.1 to PCFs of arbitrary degree d. For degree-d denominator $b(n) = \beta_d n^d + \text{l.o.t.}$, conjecture $\xi_0 = c(d)/\beta_d^{1/d}$ (Theorem 3.3): candidate $c(d) = d$ (linear), proven at $d = 2$ (Theorem 3.1), verified at $d = 4$ with spread 0 in PCF2-SESSION-Q1 (Theorem 3.2, [18]); $d = 3$ verification deferred to op:xi0-d3-direct. The earlier v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$ is empirically falsified at $d = 4$ (Theorem 3.4). *Tractability.* Resolved at $d = 2, 4$; remaining work is the direct $d = 3$ test (op:xi0-d3-direct) and a general-d Newton-polygon proof.

Open problem 9.4 (op:xi0-d3-direct)**. Status (v1.3):** Carried forward unchanged. Tractability re-checked at 1–2 hours of computation, no halt.

Carry out the direct Newton-polygon test of Theorem 3.3 at $d = 3$: select one cubic representative per Galois bin of PCF-2 v1.1 [19] ($+_{S_3}$ real, $+_{C_3}$ real, $-_{S_3}$ CM), extract the characteristic polynomial of the irregular singular point of $L_3 f = (1 - zB_3(\theta+1) - z^2)f$ at $z = 0$, measure ξ_0 at dps ≥ 80, and compare with the prediction $c(3) = 3$, i.e. $\xi_0 = 3/\beta_3^{1/3}$. Evidence at $d = 4$ has $c(4) = 4$ with spread 0 (Theorem 3.2, [18]). *Tractability.* 1–2 hours of computation along the PCF2-SESSION-Q1 `session_q1_newton.py` pattern, adapted to the cubic operator.

Open problem 9.5 (op:bridge-witness-tier)**.** For each degree-2 PCF in the catalogue, determine whether the bridge identity is B1 (β_2 only), B2 (β_2 plus one secondary), or B3 (full tuple). Currently V_{quad} and Q_{L15} share $\beta_2 = 3$ and would lie in B1's candidate set if their Painlevé labels matched; they do not (K-SCAN artefact, Theorem 3.6), pushing the V_{quad} witness toward B2 or higher. *Tractability.* Hinges on op:cc-monodromy-RH.

(v1.3): Tier discrimination is now also informed by the umbrella's $\|\Delta\|_{\text{Pet}}$ axis [28]; for the cubic catalogue the modular-discriminant signal is at $\rho = +0.638$ ($n = 50$, Bonferroni $p = 8.6 \times 10^{-6}$), suggesting a (β_d, $\|\Delta\|_{\text{Pet}}$, ξ_0)-coordinatised refinement of the B1/B2/B3 tiering.

Open problem 9.6 (op:coefficient-tuple-functoriality)**.** Make the functoriality statement of Theorem 6.1 precise: define the category of admissible PCF coefficient tuples and morphisms (rescalings, polynomial reparametrisations) and prove (or disprove) that the bridge map Φ_b is a functor. Currently stated only conjecturally. *Tractability.* Multi-year.

Open problem 9.7 (op:no-go-proof)**.** Prove or refute Theorem 5.1. *Tractability.* 1–3 sessions for a refutation if a moduli of perturbation directions ω produces a stable $L(t)$ Painlevé fit; multi-year for a proof.

Open problem 9.8 (op:cc-monodromy-RH)**.** Extend the Newton-polygon / Birkhoff formal-solution extractor of [21] to a full Riemann–Hilbert monodromy datum at the irregular singular point $z = 0$

via Stokes-matrix computation. This is the prerequisite for a rigorous Painlevé-class verdict on the five non-V_{quad} families at any finite-digit threshold. *Tractability.* 1–2 sessions per family.

Open problem 9.9 (op:channel-moduli)**.** Is the channel space **Channel** a moduli space (with positive-dimensional families of channels) or a discrete / finite set? *Tractability.* Multi-year. Likely requires the Stokes-matrix / Frobenius-manifold framework of [2].

Open problem 9.10 (op:borel-channel)**.** Is the BoT channel of Theorem 2.3 ANOMALOUS in the formal sense (the Borel-resummation section S on the BoT trans-series space does not exist for $\Delta < 0$ degree-2 PCFs), or merely under-resolved at $K \leq 24$? The CC-PIPELINE-F UF-1 finding [20] – that the BoT space and the CC space disagree at the level of the Borel singular structure – argues for the former; the K-SCAN drift of [24] is consistent with either reading. PCF-1 v1.3 [16] carries this as "*BOREL CHANNEL CONFIRMED ANOMALOUS*". *Tractability.* Bound to op:cc-formal-borel.

Open problem 9.11 (op:cc-channel-existence (legacy))**.** For families with $\deg b \geq 3$ or $\Delta \geq 0$, does the generating-function ODE (1) (or its generalisation) still admit a Newton-polygon Birkhoff formal pair, i.e., does the CC channel exist as a non-empty triple (D, T, S)? *Tractability.* Joint with op:xi0-degree-d.

(v1.3): The cubic case is now bound to PCF-2 v1.3 [28] (Conjecture B5/B6 in their $d = 3$ restricted form), which gives a structural footprint to test in the cubic CC channel.

Open problem 9.12 (op:painleve-classification)**.** For families admitting a CC formal pair, classify the attainable Painlevé labels. Empirically, only $P_{III}(D_6)$ appears in the present roster (one family). *Tractability.* Bound to op:cc-monodromy-RH.

Open problem 9.13 (op:non-piii (legacy))**.** Are there degree-2 $\Delta < 0$ PCFs whose CC reduction is non-P_{III}? PCF-1 v1.3 Sec. 8 redirects this to op:borel-channel on the grounds that QL06 / P-V was ruled out by Session D and that the Borel channel diagnoses the candidate set is artefact-dominated. *Tractability.* Bound to op:cc-monodromy-RH.

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The v1.3 revision additionally absorbs the SIARC umbrella v2.0 invariant-triple framing [27], the PCF-2 v1.3 cubic-modular Petersson-height result [28], and the Theory-Fleet H4 median-resurgence prediction [30]. AI tools (GitHub Copilot powered by Anthropic Claude Opus 4.7, Anthropic Claude) were used for prose drafting, consistency-checking, and structural re-organisation; all conjectures, theorem statements, and editorial decisions remain the author's. All numerical claims trace to exact computations stored under `sessions/2026-05-{01,02}/` on the SIARC bridge with AEAL claim entries in `claims.jsonl`.

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($\sim 30\times$ improvement in Bonferroni p). Phase B/C correlation table used as the empirical anchor of Sec. 3.6 of this outline. B. Direct empirical cross-reference.

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